

INTERNATIONAL news **US tightens export restrictions on China and China bans US chips in government systems**

The US has tightened the regulations to make it more difficult for China to obtain US artificial intelligence (AI) chips and chip making tools. The new rules, effective from April 4, 2024, extend the restrictions to cover laptops containing those chips. China has announced new procurement reforms that urge government institutions to shift from US tech giants like Intel and AMD to domestic alternatives, putting \$74 billion of American tech revenue in jeopardy. The reforms are part of the 'XinChuang' or 'IT application innovation' national strategy, aimed at achieving technological self-reliance in state matters. State-owned enterprises have been directed to switch to domestic technology providers by 2027.

US and EU to explore semiconductor disruptions, focusing on legacy chips from China

The US and EU have agreed to extend their collaboration for three more years to address disruptions in the semiconductor sector, focusing on legacy chips from China. China accounts for around 60% of global legacy chip production. They aim to share market intelligence on non-market practices in China that might have been causing supply chain distortions. The partnership will also explore alternatives to harmful chemicals used in chip manufacturing.

The Dutch government asked to restrict servicing of ASML equipment in China

The US government in a recent meeting pushed the Dutch government, which is a strong US ally, to restrict ASML from servicing some of its equipment in China. Advanced Semiconductor Materials Lithography (ASML) is the world's biggest chip-making equipment manufacturer and China is its second largest market, after Taiwan. ASML's equipment is almost impossible to replace, and no one has tried to use them without the company's maintenance.

Intel gets \$20 billion in grants and loans to boost domestic semiconductor production

US awarded Intel nearly \$20 billion in grants and loans to boost domestic semiconductor production, marking the largest US government investment in chip manufacturing. In response to this Intel is planning to spend \$100 billion across four US states in five years to build and expand factories and hopes to secure an additional \$25 billion in tax breaks. Intel plans to transform empty fields near Columbus, Ohio, into the world's largest AI chip manufacturing site, starting as early as 2027. It includes revamping sites in New Mexico and Oregon and expanding operations in Arizona.

Infineon and Amkor partner to boost chip manufacturing in Europe

Infineon Technologies has partnered with Amkor Technology, Inc. to operate a dedicated packaging and test centre at Amkor's manufacturing site in Porto, Portugal, with operations set to begin in the first half of 2025. This collaboration marks a departure from the traditional Outsourced Semiconductor Assembly and Test (OSAT) model. Amkor will accommodate the production line and provide dedicated clean room space, while Infineon will supply an onsite team for engineering and development support.

Keysight acquires Spirent Communications

Test and measurement equipment giant, Keysight Technologies, outbid Viavi Solutions to acquire Spirent Communications for around \$1.46 billion. Spirent's expertise in automation and software-based next-generation communication networks, including 6G, is a key

driver behind the acquisition. Keysight expects new serviceable available market opportunities of up to \$1.5 billion following the acquisition of Spirent.

Microsoft to bring NVIDIA's gen AI tech to Azure

Microsoft Corp. and NVIDIA have partnered to integrate NVIDIA's generative AI and Omniverse technologies across various Microsoft platforms, including Azure, Azure AI services, Microsoft Fabric, and Microsoft 365. The collaboration includes bringing Grace Blackwell processor to Azure and integrating DGX Cloud with Microsoft Fabric. NVIDIA GPUs and Triton Inference Server will also power AI inference predictions in Microsoft Copilot for Microsoft 365.

S&P 500 to include Supermicro Computer

Supermicro Computer is set to join the Standard & Poor's (S&P) 500, a benchmark for the overall performance of the US stock market. Supermicro has a unique advantage among server makers due to its close ties with NVIDIA. It has historically been among the first to receive AI chips from NVIDIA and AMD, allowing it to launch products faster than rivals Dell and Hewlett Packard Enterprise. The company's stock has outperformed even NVIDIA, with its market value reaching \$60 billion, up from \$5 billion before the launch of ChatGPT.

UXL to challenge NVIDIA's dominance in AI chips

The UXL Foundation, a consortium of big techs that aims for an open standard accelerator software ecosystem, now aims to challenge NVIDIA's AI dominance. The consortium plans to develop an open source suite of software and tools to power various AI accelerator chips, an area currently dominated by NVIDIA's CUDA software platform. UXL will finalise technical specifications in the first half of the year, with plans to mature the details by year-end.

TSMC and Synopsys integrate NVIDIA's cuLitho in chip manufacturing

TSMC and Synopsys have integrated NVIDIA's computational lithography platform cuLitho with their software and manufacturing processes. The companies are going into production with cuLitho to speed up the manufacturing of advanced semiconductor chips. This integration aims to accelerate chip fabrication and support future Blackwell GPUs. NVIDIA has introduced new generative AI algorithms that enhance cuLitho, speeding up the manufacturing process. These algorithms help create a near-perfect inverse mask for optical proximity correction, doubling the speed of this process.

Apple to make robotic personal assistants

Apple is working on developing personal robots. The development comes shortly after the company abandoned its electric car project. The tech giant is working on a robot designed to assist people at home by following them around. Apple has been exploring new sectors for revenue inflow beyond its core iPhone businesses.

Microsoft and Quantinuum achieve most reliable qubit generation till date

Microsoft and quantum computing company, Quantinuum, have achieved the most reliable qubit generation till now. This development is part of the ongoing competition among tech companies like Microsoft, Google, and IBM. Microsoft applied its error-correction algorithm to Quantinuum's physical qubits. Microsoft plans to release this technology to its cloud computing customers soon. This breakthrough could potentially shorten the timeline to achieve stable qubits by at least two years or more.